SEUNGPIL LEE

(+82) 10 3283 0369 ⋄ iamseungpil@gm.gist.ac.kr
15 Gwacheon-daero 8-gil, Gwacheon-si
Gyeonggi-do, Republic of Korea, 13824

Personal Statement

Master's researcher who moves in both directions between AI and the human mind: from AI back to the mind it imitates, and from the mind forward into the AI it shapes. My current focus is the cognition of intelligent systems, studied through LLMs — mechanistic interpretability keeps me honest about mechanisms, and memory-augmented agents that learn at test time are the current testbed. Coming in through literature and philosophy of mind, I draw on cognitive science and STS alongside the engineering.

EDUCATION

Gwangju Institute of Science and Technology (GIST) — *March 2025 – present*
Master's Program, Major in AI Convergence

Gwangju Institute of Science and Technology (GIST) — *March 2018 – December 2024*
B.S. in Electrical Engineering and Computer Science — GPA: 3.66/4.50
Minor in Mathematics, Minor in Literature

UC Berkeley — *June 2019 – August 2019*
Berkeley Summer Session Program — GPA: 4.00/4.00

PUBLICATIONS

Journal Articles

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sanha Hwang, Wongyu Seo, Jiwon Park, Seokki Lee, Sejin Kim, and Sundong Kim. "Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus." *ACM Transactions on Intelligent Systems and Technology* (**ACM TIST**), 2024.
- Sanha Hwang, **Seungpil Lee**, Sejin Kim, and Sundong Kim. "Solution Augmentation for ARC-AGI Problems Using GFlowNet: A Probabilistic Exploration Approach." *Transactions on Machine Learning Research* (**TMLR**), 2025.

Conference Papers

- **Seungpil Lee***, Donghyeon Shin*, Klea Lena Kovacec, and Sundong Kim. "From Generation to Selection: Findings of Converting Analogical Problem-Solving into Multiple-Choice Questions." *Findings of EMNLP* (**EMNLP Findings**), 2024.
- Hosung Lee*, Sejin Kim*, **Seungpil Lee**, Sanha Hwang, Jihwan Lee, Byung-Jun Lee, and Sundong Kim. "ARCLE: The Abstract and Reasoning Corpus Learning Environment for Reinforcement Learning." *Conference on Lifelong Learning Agents* (**CoLLAs**), 2024.

Preprints

- Heejun Kim*, **Seungpil Lee***, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, and Sundong Kim. "From Noise to Diversity: Random Embedding Injection in LLM Reasoning." *arXiv preprint*, 2026.
- **Seungpil Lee**, Donghyeon Shin, Yunjeong Lee, and Sundong Kim. "Can Large Language Models Develop Gambling Addiction?" *arXiv preprint*, 2025.

Workshop Papers

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sejin Kim, and Sundong Kim. "Reasoning Abilities of Large Language Models through the Lens of Abstraction and Reasoning." *NeurIPS Workshop on System-2 Reasoning at Scale*, 2024.
- Donghyeon Shin*, **Seungpil Lee***, Klea Lena Kovacec, and Sundong Kim. "Regulation Using Large Language Models to Generate Synthetic Data for Evaluating Analogical Ability." *IJCAI Workshop on Analogical Abstraction*, 2024.

Domestic Publications

- **Seungpil Lee**, Jihwan Lee, and Sundong Kim. "Evaluating Prior Knowledge of ARC Using World Models." *Korea Software Congress*, 2023.
- Jihwan Lee, **Seungpil Lee**, Sejin Kim, and Sundong Kim. "Extracting the Core Knowledge of ARC with the World Model." *Korea Software Congress*, 2023.
- Donghyeon Shin, Sanha Hwang, Seokki Lee, Yunho Kim, **Seungpil Lee**, and Sundong Kim. "MC-LARC Benchmark to Measure LLM Reasoning Capability." *Korea Software Congress*, 2023.

* indicates equal contribution

SKILLS

Programming	C, C++, Java, JavaScript, Python
Tools	LaTeX, Spring Boot, PyTorch, Hugging Face
Language	Korean (Native), English (Intermediate)

EXPERIENCE

Microsoft Research Asia (MSRA), Shanghai — *March 2026 – August 2026*
Research Intern, Collaborative Research Program

- Designing structured memory frameworks for self-evolving agents that update their knowledge at inference time without parameter retraining
- Studying how test-time memory generation enables compositional generalization on reasoning tasks where vanilla LLMs fail to reuse learned primitives
- Bridging mechanistic analysis of reasoning failures with memory-based mitigations

DataScience Lab, GIST — *September 2023 – present*
Undergraduate Intern → Master's Student

- LLM evaluation with cognitive science tools (Language of Thought Hypothesis)
- LLM alignment using RLVR methods; Mechanistic interpretability with Sparse Autoencoders

Teaching Assistant, Computer Networking — *September 2023 – December 2025*

- Designed assignments connecting theory to implementation; SDN demo with Ryu and Mininet

BioComputing Lab, GIST — *June 2020 – January 2021*
Undergraduate Intern

- Developed Spiking Neural Network methods with synthetic gradients

AWARDS & FUNDING

National Research Foundation of Korea (NRF) Funding — *Sep 2025 – Sep 2026*
Research funding for outstanding master's students

Best Undergraduate Thesis Award, GIST EECS — *February 2025*
Awarded for graduation thesis on MC-LARC benchmark

Korean Government Scholarship, GIST College *March 2018 – August 2024*
Full scholarship for students at GIST

Scholarship for Summer Session Abroad *June – August 2019*
UC Berkeley Summer Session scholarship

Navy AI Competition *August 2023*
Award in drone object detection competition

K-StartUp Finals *November 2023*
Advanced to finals as ML researcher and back-end developer

EXTRA-CURRICULAR

Published Author – Short Story Collection *2019*
Published a 400-page short story collection exploring consciousness and identity

AGIST – Deep Learning Study Group *August 2020 – January 2021*
Presentations on Spiking Neural Networks with STDP learning